MIL-F-81739(AS)
15 August 1969
Superseding
AS-1179A
23 June 1969

MILITARY SPECIFICATION

FUZE FLARE MK 364 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the manufacture, assembly and preparation for delivery of the Fuze, Flare Mk 364 Mod 0 and the methods of examination and tests upon which product acceptance shall be based. The fuze is a variable time delay type with a glass filled polystyrene body. The ejection delay selection is 500 to 14,000 feet of fall. The expelling charge consists of a black powder pellet. The fuze is classified as Class B Explosive (Code of Federal Regulations 49 CFR 71-78) and Class IIC Explosive (Coast Guard Rules and Regulations for Military Explosives and Hazardous Munitions CG 108).

1.2 Definitions. The meaning of terms used in this specification shall be as listed in 6.4.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-45662	Calibration System Requirements

FSC-1370

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1314	Safety Precaution for Explosive Loaded Items

DRAWINGS

Naval Air Systems Command

PL 2127860	Fuze, Flare Mk 364 Mod 0

(And all PL's, drawings, specifications, standards, and publications listed thereon.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Interstate Commerce Commission Rules and Regulations for Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite the latest edition and supplement.)

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COAST GUARD RULES AND REGULATIONS

CG 108 Rules and Regulations for Military Explosives and Hazardous Munitions

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Inspection and manufacture. The fuze shall be manufactured, examined, tested and packed in accordance with contract documents, Parts Lists, PL 2127860, and this specification.

3.1.1 Preproduction sample. An acceptable preproduction sample of fuzes shall be furnished by the contractor. The fuzes, their components and all materials shall be of single type, grade, class, size and composition and shall comply with all applicable provisions of this document. The sample shall be manufactured using the proposed full production methods. The preproduction sample shall not be applied as part of the quantity specified for delivery by the contract or order.

3.2 Examinations.

3.2.1 Components, assemblies and materials. All components, assemblies and materials of the fuze shall meet all the requirements of their respective specifications or drawings.

3.2.1.1 Primer. The primer shall meet the requirements of the applicable drawing.

3.2.2 Fuze. The fuze shall meet the requirements of this specification.

3.3 Performance and product characteristics. The fuze shall meet the acceptance criteria for performance and product characteristics in the sequence indicated in Table II.

3.3.1 Performance.

3.3.1.1 Sealing. The fuze shall withstand a vacuum of 6.0 ± 1.0 inches of mercury (24 inches of mercury absolute pressure) for a minimum period of 60 seconds without signs of leakage.

3.3.1.2 Static functioning. The fuze shall meet the following

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requirements:

a. Shall not fire the expelling charge less than 1.5 seconds after pulling lanyard.

b. Function (shall rupture ejection charge cover).

c. Ejection delay time: Time from pulling lanyard to rupturing of the ejection charge cover shall be within the tolerance specified in Table I.

TABLE I

FOR STATIC TEST EJECTION DELAY TIMES

Feet of Fall	Seconds*	Feet of Fall	Seconds*
500	5.6 ± 2.0	8,000	31.1 ± 6.0
1,000	8.1 ± 3.0	9,000	33.4 ± 6.0
2,000	12.6 ± 3.0	10,000	35.7 ± 7.0
3,000	16.5 ± 4.0	11,000	37.8 ± 7.0
4,000	20.0 ± 4.0	12,000	39.9 ± 8.0
5,000	23.2 ± 4.0	13,000	42.0 ± 8.0
6,000	26.1 ± 5.0	14,000	44.1 ± 8.0
7,000	28.7 ± 5.0		

*For test locations from 0 to 1500 above MSL.

3.3.2 Durability and environmental.

3.3.2.1 Jolt. The fuze shall withstand jolt without exploding, burning or coming apart.

3.3.2.2 Jumble. The fuze shall withstand jumble without exploding, burning or coming apart.

3.3.2.3 Forty (40) foot drop. The fuze shall withstand forty (40) foot drop without exploding or burning.

3.3.2.4 Aircraft vibration. The fuze shall withstand aircraft vibration without exploding, burning or coming apart.

3.3.2.5 Low temperature. The fuze shall be conditioned at a temperature of -65°F ± 5°F for a minimum period of 24 hours without exploding or burning.

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3.3.2.6 Safety functioning. The fuze shall not explode, deflagrate, or rupture the ejection charge cover less than 1.5 seconds from when the lanyard is pulled.

3.4 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents employed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Preproduction sample. The contractor shall deliver a sample of 180 fuzes to the destination specified (see 6.2) for examination, testing, evaluation and acceptance as specified in Table II, as applicable. Any production of the item prior to authorization by the procuring agency shall be at the contractor's risk. The sample shall be examined as specified in 4.2 and prepared for delivery as specified in Section 5.

4.1.2 Production lot. A production lot shall consist of 3201 to 10,000 fuzes offered for delivery at one time including test samples. The test samples shall not be applied as part of the quantity specified for delivery by the contract or purchase order. Each production lot shall contain only primers produced during one year by one manufacturer. Sampling, examination, testing and acceptance of production lots shall be performed as specified in the steps given below. Upon rejection of any lot, the next lot, in addition to the applicable plan, shall be examined, tested and accepted for the requirement(s) which caused the rejection, unless the requirement(s) is included in the applicable plan.

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STEP 1. Upon starting production or delivery of an acceptable pre-production sample, as applicable, the first two production lots shall be examined, tested in sequence and accepted in conformance with Table II, Plan A.

STEP 2. Production lots other than those defined in Step 1 shall be examined, tested in sequence and accepted in conformance with Table II, Plan B, except that one lot shall be randomly selected from each five consecutive production lots and examined, tested in sequence and accepted in conformance with Table II, alternately selecting Plan A and Plan C.

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TABLE II
INSPECTION PLANS

PLAN [1]				SECTION REFERENCE	EXAMINATIONS AND TESTS	PREPRODUCTION SAMPLING PLANS		PRODUCTION LOT SAMPLING PLANS	
PRE-PROD	A	B	C		1 2 3	SAMPLE SIZE	ACCEPTANCE CRITERIA [2]	SAMPLE SIZE	ACCEPTANCE CRITERIA [2]
X	X	X	X	4.2.2	Examination	180 Fuzes	Critical Ac 0 Re 1 Major Ac 5 Re 6 Minor Ac 10 Re 11	See Para 4.2.2	See Para 4.2.2
X			X	4.3.2.1 3.3.2.1	Jolt Test	13 Fuzes	Ac 0 Re 1	MIL-STD-105 Inspection Level S-1	Ac 0 Re 1
				4.3.2.6 3.3.2.6	Safety Functioning Test		Ac 0 Re 1		Ac 0 Re 1
X			X	4.3.2.2 3.3.2.2	Jumble Test	13 Fuzes	Ac 0 Re 1	MIL-STD-105 Inspection Level S-1	Ac 0 Re 1
				4.3.2.6 3.3.2.6	Safety Functioning Test		Ac 0 Re 1		Ac 0 Re 1
X	X		X	4.3.2.3 3.3.2.3	Forty (40) Foot Drop Test	13 Fuzes	Ac 0 Re 1	MIL-STD-105 Inspection Level S-1	Ac 0 Re 1
				4.3.2.6 3.3.2.6	Safety Functioning Test		Ac 0 Re 1		Ac 0 Re 1
X	X		X	4.3.2.4 3.3.2.4	Aircraft Vibration Test	32 Fuzes	Ac 0 Re 1	MIL-STD-105 Inspection Level S-3	Ac 0 Re 1
				4.3.1.2 3.3.1.2	Static Functioning Test		(a) Ac 0 Re 1 (b) Ac 2 Re 3 (b&c) Ac 3 Re 4		(a) Ac 0 Re 1 (b) AQL 2.5 (b&c) AQL 6.5
X	X		X	4.3.1.1 3.3.1.1	Sealing Test	32 Fuzes	Ac 2 Re 3	MIL-STD-105 Inspection Level S-3	AQL 6.5
				4.3.2.5 3.3.2.5	Low Temperature Test		Ac 0 Re 1		Ac 0 Re 1
				4.3.1.2 3.3.1.2	Static Functioning Test		(a) Ac 0 Re 1 (b) Ac 2 Re 3 (b&c) Ac 3 Re 4		(a) Ac 0 Re 1 (b) AQL 2.5 (b&c) AQL 6.5
X	X	X	X	4.3.1.1 3.3.1.1	Sealing Test	50 Fuzes	Ac 7 Re 8	MIL-STD-105 Inspection Level S-4	AQL 6.5
				4.3.1.2 3.3.1.2	Static Functioning Test		(a) Ac 0 Re 1 (b) Ac 3 Re 4 (b&c) Ac 7 Re 8		(a) Ac 0 Re 1 (b) AQL 2.5 (b&c) AQL 6.5
X				4.3.2.7	Grouped Data Acceptance	114 Fuzes	(b) Ac 6 Re 7 (b&c) Ac 13 Re14	None	None

NOTES:

1 X - Denotes tests applicable to plan
2 Ac - Acceptance number
Re - Rejection number

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4.1.3 Test equipment and gages. All test equipment and gages shall be supplied and maintained by the contractor as specified in Specification MIL-I-45607. The contractor shall maintain a calibration system as specified in Specification MIL-C-45662.

4.1.3.1 Test conditions. The maximum allowable tolerances on test conditions, unless otherwise specified, shall be as follows:

1. Temperature: Plus or minus 3°C, 5.4°F.

2. Relative humidity: Plus 5 percent, minus 0 percent.

3. Vibration amplitude: Plus or minus 10 percent.

4. Vibration frequency: Plus or minus 2 percent.

4.1.4 Tests and examinations classification. The tests and examinations herein are classified in regard to relative importance as Critical, Major or Minor. Tests and examinations classified as Critical are identified by the symbol (C) and Major by (M). The number following the symbol indicates the serial number of the test or examination. Tests which are not annotated are classified Minor.

4.2 Examinations.

4.2.1 Components, assemblies and materials examination. Prior to assembly, all components, assemblies and materials used in the fuze shall meet the requirements of 3.2.1.

* 4.2.1.1 Primer examination (M101). In the absence of previous inspection of the primer, inspection shall be performed as specified on Drawing 2150846, 2151306 or 7645336-2 as applicable.

4.2.2 Fuze examination (M102). Examinations shall be made to ascertain that the fuzes meet the requirements of drawings listed on PL 2127860. Only fuzes containing no defective classified characteristics shall be used for further testing or returned to the lot.

4.2.3 Packaging, packing and marking examination (C1). The packing of the fuze and marking of the pack shall conform to Section 5 of this specification.

4.3 Tests. Random samples of fuzes shall be selected and tested as specified in 4.1, as applicable.

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4.3.1 Performance test.

4.3.1.1 Sealing test (M103). The fuze shall be tested in a water tank equipped for observation under vacuum. The water temperature shall be 45 degrees F minimum to 75 degrees F maximum.

CAUTION: The water tank should be equipped so that the fuze must be raised above the water level while still under vacuum before releasing the vacuum and removing the fuze.

The uppermost surface of the fuze shall be submerged 4 to 6 inches below the water surface and subjected to a vacuum of 6.0 ± 1.0 inches of mercury (24 inches of mercury absolute pressure) for a minimum period of 60 seconds. Leakers are indicated by air bubbles issuing from the fuze. Do not mistake the escape of occluded air for leakage. Defectives are fuzes failing to meet the requirements of 3.3.1.1.

4.3.1.2 Static functioning test (C2). The fuze shall be tested by placing the fuze on a fireproof surface of sufficient size to guard against fire hazards in normal burning and in case of explosive burning. The fuze delay time shall be set at values randomly selected. The fuze shall be held in position to prevent movement during functioning. The fuze shall be initiated by a quick pull on the lanyard with an extension of sufficient length so personnel can be at a safe distance or behind a shield. Record ejections of less than 1.5 second after pulling lanyard, functioning and ejection delay times. Defectives are fuzes failing to meet the requirements of 3.3.1.2.

4.3.2 Durability and environmental tests.

4.3.2.1 Jolt test (C3). The fuze shall be subjected to the jolt test prescribed in Test 101 of Standard MIL-STD-331. Defectives are fuzes failing to meet the requirements of 3.3.2.1.

4.3.2.2 Jumble test (C4). The fuze shall be subjected to jumble test prescribed in Test 102 of Standard MIL-STD-331. Defectives are fuzes failing to meet the requirements of 3.3.2.2.

4.3.2.3 Forty (40) foot drop test (C5). The fuze shall be subjected to the forty (40) foot drop test prescribed in Test 103 of Standard MIL-STD-331, except the fuze shall be dropped free-fall without guidance system or associated equipment. Defectives are fuzes failing to meet the requirements of 3.3.2.3.

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4.3.2.4 Aircraft vibration test (C6). The fuze shall be subjected to the aircraft vibration test prescribed in Specification MIL-E-5272, Vibration Procedure XII, at ambient temperature only. The fuze shall be assembled to the vibration table using a suitable mounting. Defectives are fuzes failing to meet the requirements of 3.3.2.4.

4.3.2.5 Low temperature test (C7). The fuze shall be conditioned in a suitable chamber at a temperature of -65°F ± 5°F for a minimum period of 24 hours and shall be static functioned while at that temperature. Defectives are fuzes failing to meet the requirements of 3.3.2.5.

4.3.2.6 Safety functioning test (C8). The fuzes that are damaged and appear unsafe to function shall not be tested. Testing shall be in accordance with 4.3.1.2. Record fuzes which explode, deflagrate or rupture the ejection charge cover less than 1.5 seconds from when the lanyard is pulled. Defectives are fuzes failing to meet the requirements of 3.3.2.6.

4.3.2.7 Grouped data acceptance. Data shall be grouped for the static functioning test on the sample of 114 fuzes composed from the following groups:

Group (a)	32 fuzes from the aircraft vibration test of 4.3.2.4
Group (b)	32 fuzes from the sealing-low temperature test of 4.3.2.5
Group (c)	50 fuzes from the sealing test of 4.3.1.1

Acceptance shall be as specified in Table I.

5. PREPARATION FOR DELIVERY

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5.1 Packing.

5.1.1 Level A. In addition to the following, Level A packing shall be in accordance with the Code of Federal Regulations 49 CFR 71-78 for Class B Explosives.

5.1.1.1 Thirty (30) fuzes shall be packed in accordance with Drawing 2127817 unless otherwise specified (see 6.2).

5.2 Marking.

5.2.1 Special marking. In addition to the requirements of the contract or order, the outer shipping container shall be marked in accordance with the Code of Federal Regulations 49 CFR 71-78 for Class B Explosives. (Special Fireworks)

5.2.2 Normal marking. In addition to any requirements of the contract or order, packs shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The Fuze, Flare Mk 364 Mod 0 is intended for use as a component of the Flare, Aircraft, Parachute Mk 45 Mod 0.

6.2 Ordering data. Procurement documents shall specify:

a. Title, symbol and date of this specification,

b. Quantity of fuzes being ordered,

c. Test facility where preproduction samples are to be shipped for evaluation,

d. QE Laboratory coordinating surveillance,

e. Data information required (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraph below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless justified and specified in the contract or order. The data to be furnished shall be as listed on DD Form 1423, which shall be attached to and made a part of the contract or order. NAVWEPS

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Form 4200/15 shall be attached where applicable.

6.3.1 The contractor shall prepare 5 inch X 8 inch data cards made of white index bristol board approximately 110 lbs. ream weight in accordance with Figure 1 for each lot, if applicable (see 6.2). The contractor shall fill in all required information, including information for the following components:

Component

Primer

6.3.2 The government inspector shall endorse the cards and distribute one each as follows:

a. Procuring activity,

b. SPCC (Code 730, Mechanicsburg, Pa., if different than the procuring activity),

c. QE Laboratory coordinating surveillance,

d. The designated test activity (see 6.2).

6.4 Definitions. The meaning of the terms used in this specification shall be as listed below.

6.4.1 Fuze. The term "fuze" shall mean "Fuze, Flare Mk 364 Mod 0" as depicted on Drawing 2127860.

6.5 Conflicting requirements. Conflicting requirements arising between this specification and any other specification, publication or drawing listed herein shall be referred in writing to the procuring agency for clarification.

* 6.6 Safety precautions. The safety precautions specified in MIL-STD-1314 shall be observed during loading, assembly, testing and handling operations. The use of this standard shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. The observation of these safety precautions is mandatory at government plants.

* 6.7 Supersession data. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this

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document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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REMARKS

ACCEPTANCE DATA

	LOCATION	DATE OF TEST	NO. OF SAMPLES	TEST REPORT REFERENCE
ORIGINAL TEST ▶				
RETEST ▶				

REMARKS: *(Note all defects noted. Describe rewarking, if any. Use top of card if necessary)*

DATE RELEASED	AUTHORITY FOR RELEASE	INSPECTOR

PYROTECHNIC AMMUNITION MANUFACTURING DATA CARD (BACK) NAVORD FORM 1294 (10-44)

(Name of Pyrotechnic Ammunition) *(Mk.—Mod. and color)*	*(Contract No.)*	*(Assembly Lot No.)*
MANUFACTURER	NUMBER IN LOT	DATE OF MANUFACTURE

SPECIFICATIONS	DRAWING NOS. *(List revisions actually used.)*

HOW PACKED

COMPONENT	MANUFACTURED BY	LOT NO.	DATE OF MFG.	ADDITIONAL DATA

REMARKS: *(List below all deviations in this lot from specifications and drawings with all references to authorizations for changes and waivers. All changes in procedure, sources of material, or unusual circumstances during manufacture should be noted. Where alternates are allowed, specify which is used. Use back of card if necessary.)*

PYROTECHNIC AMMUNITION MANUFACTURING DATA CARD NAVORD FORM 1294 (10-44)

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFY

MIL-F-81739(AS) FUZE FLAIR MK 364 MOD 0

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding Officer

U. S. Naval Ammunition Depot (Code RD)

Crane, Indiana 47522

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